Six-Level Tetrahedral Rhythm ⇄ Hologram Moonshine — Integration Note (v1.0)

Status: Integration-ready specification and proof-obligations.
Scope: Defines the minimal, certifiable conditions under which a Six-Level Tetrahedral Rhythm integrates with the Hologram Moonshine (Φ-Atlas) stack.
Keywords: tetrahedral rhythm, Φ-map, C768 schedule, R96 resonance, equivariance, orbit reduction, audit bundle.

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0) Notation & Setup

- Integers mod n : $\mathbb{Z}/n\mathbb{Z}$. Bitwise XOR is denoted $\oplus$ on fixed-width binary registers.
- **Pages**: $P = \mathbb{Z}/48\mathbb{Z}$, **Bytes**: $B = \mathbb{Z}/256\mathbb{Z}$.
- **Boundary lattice**: $G = P \times B$, hence $|G| = 48 \cdot 256 = 12{,}288$.
- **Anchors**: $S = \{(8m, 0) : m = 0, 1, 2, 3, 4, 5\} \subseteq G$. (Six evenly spaced page anchors.)
- **Reference subgroup**: $U_{\text{ref}} \cong (\mathbb{Z}/2\mathbb{Z})^{11}$ acting on G by toggling a fixed generating set of XOR-masks: three page masks and eight byte masks. The action is written $u \cdot (p, b)$.
  Remark: page masks act on the 6-bit register of $p \in [0, 47]$ (with wrap behavior defined by the register↔mod map), byte masks on the 8-bit register of $b \in [0, 255]$.

Throughout, **deterministic configuration** refers to the fixed choice of (i) anchor set S , (ii) generator order for U_{ref} , (iii) hash salt and (iv) label seeds—bundled in the audit artifacts (see §5).

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1) Boundary Substrate & Subgroup Certificate

Requirement 1.1 (Group and action). Implement $G = P \times B$ and an explicit action of $U_{\text{ref}} \cong (\mathbb{Z}/2)^{11}$ via XOR-masks as above.

Requirement 1.2 (Order and freeness). Prove/verify: - $|U_{\text{ref}}| = 2^{11} = 2048$. - The action of U_{ref} is **free on each anchor** $s \in S$: if $u \cdot s = s$ then $u = \text{id}$.

Requirement 1.3 (Tiling certificate). Show that the six orbits $\mathcal{O}(s) = U_{\text{ref}} \cdot s$ are pairwise disjoint and cover G . Consequently,

$$|\mathcal{O}(s)| = 2048 \quad \text{and} \quad 6 \times 2048 = 12{,}288 = |G|.$$

Proof sketch. Freeness gives $|\mathcal{O}(s)| = |U_{\text{ref}}|$. Disjointness follows because any intersection would contradict freeness via anchor offsets. Cardinality then forces the cover.

Deliverable. A machine-checkable *subgroup certificate* stating the generator set, order, freeness log on S, and the orbit-partition (tiling) witness.

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2) Canonical Fair Schedule C768 & Closure Checks

Define the **Canonical Fair Schedule**

$$\sigma_{\mathrm{CFS}} : \{0, \ldots, 767\} \to G, \qquad \sigma_{\mathrm{CFS}}(t) = (t \bmod 48,\ t \bmod 256).$$

Fairness properties. Each page appears exactly 16 times (since $768/48 = 16$), and each byte appears exactly 3 times (since $768/256 = 3$). Prefix-fairness holds for any prefix length divisible by 48 or 256.

Observable family. For affine boundary observables

$$f(p,b) = d + \alpha_p + \gamma_b,$$

require closed forms for the schedule sums

$$S^{(r)} = \sum_{t=0}^{767} f(\sigma_{\mathrm{CFS}}(t))\, t^r, \quad r \in \{0,1,2\},$$

matched exactly (integers) or to certified machine tolerances. In particular, for $f \equiv 1$:

$$S^{(0)} = 768, \qquad S^{(1)} = \sum_{t=0}^{767} t = 294{,}528, \qquad S^{(2)} = \sum_{t=0}^{767} t^2 = 150{,}700{,}160.$$

Deliverable. *C768 acceptance report* with (i) fairness counts, (ii) $S^{(r)}$ values for the chosen observable basis $\{1, \alpha_p, \gamma_b\}$, and (iii) proof/verification notes.

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3) Bulk–Boundary Φ-Map & Equivariance (Φ-Compatibility)

Let the **bulk space** be $\mathcal{B} := A \times \mathbb{Z}_2^{10}$ with $A = \{0,1,2,3,4,5\}$. Write bulk elements as (a,z) with $a \in A$ and $z \in \mathbb{Z}_2^{10}$.

Definition (Φ). Fix binary matrices $M_p \in \mathrm{Mat}_{6\times 10}(\mathbb{F}_2)$, $M_b \in \mathrm{Mat}_{8\times 10}(\mathbb{F}_2)$, and an anchor embedding $\iota : A \to \mathbb{F}_2^6$ that sends $a \mapsto$ the 6-bit register of page $8a$. Define

$$\Phi : A \times \mathbb{Z}_2^{10} \longrightarrow G\ (a,z) \quad \longmapsto (\,p, b\,) \text{ with} \Big\{ \text{page register } p_{\text{bits}} = \iota(a) \oplus M_p z,\ \text{byte register } b_{\text{bits}} = M_b z,$$

then decode registers to $p \in \mathbb{Z}/48\mathbb{Z}$ and $b \in \mathbb{Z}/256\mathbb{Z}$ via the fixed register↔mod map.

Equivariance condition. For each generator $u \in U_{\text{ref}}$ there exists $e(u) \in \mathbb{Z}_2^{10}$ (a bulk toggle) such that

$$\Phi\big(a,\ z \oplus e(u)\big) \;=\; u \cdot \Phi(a,z) \quad \text{for all } (a,z) \in \mathcal{B}.$$

Equivalently, the rows of $[M_p; M_b]$ span the boundary XOR masks, and the mapping $u \mapsto e(u)$ is a group monomorphism into $(\mathbb{Z}_2^{10}, \oplus)$ composed with Φ .

Tiling property via Φ. The six bulk points $(a, 0)$, $a \in A$, map under U_{ref} -transport through Φ to six disjoint orbits that cover G . This reproduces the tiling of §1 at the level of Φ-transport.

Deliverable. *Φ-compatibility report* including: (i) concrete M_p, M_b, ι ; (ii) a table of generators $u \mapsto e(u)$; (iii) logs showing equivariance on a spanning set of bulk states; and (iv) a proof that the Φ-transported anchor orbits partition G .

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4) R96 Resonance Classes (Deterministic Canonicalization)

We classify boundary configurations up to the U_{ref} action.

Generator order. Fix a total order on the 11 generators (e.g., page masks first in weight-ascending order, then byte masks in weight-ascending order). All canonicalization is with respect to this fixed order.

Gray-orbit reduction. Traverse each U_{ref} -orbit via Gray-code on the 11-bit toggle register. For each element, derive a *canonical label* by: 1. Computing a label vector (e.g., evaluation of selected invariants), 2. Hashing the serialized label using a fixed cryptographic hash (e.g., SHA-256 with a fixed salt), 3. Selecting the unique representative that minimizes the lexicographic order of (hash, raw label, toggle word) under the fixed generator order.

Normalization (pair-normalized unity). Apply a deterministic normalization to any unity-valued labels so that normalization commutes with the U_{ref} action. This pins canonical reps across isomorphic unity relabelings.

Acceptance invariant. The canonical enumeration must yield exactly **96 resonance classes** with multiplicity multiset

$$\{\,2:32,\ 3:64\,\},$$

meaning 32 classes occur with multiplicity 2 and 64 classes with multiplicity 3 under the specified selector/normalizer.

Deliverable. *R96 report* including: (i) the canonicalization configuration (generator order, salt), (ii) the class list with multiplicities, and (iii) certificates that normalization commutes with group action and is invariant under unity relabelings.

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5) Determinism & Auditability (Audit Bundle)

Integration is **reproducible** iff the following machine-readable bundle is provided: 1. **Config manifest**: anchors S , generator order, hash salt, labeling seed(s). 2. **Schedule digest**: hash and summary of σ_{CFS} and fairness counts. 3. **Subgroup certificate**: generator set, order 2^{11} , freeness log on S , tiling witness. 4. **Φ-compatibility logs**: matrices M_p, M_b , anchor embedding ι , generator-to-bulk map $u \mapsto e(u)$, equivariance checks, Φ-tiling witness. 5. **R96 enumeration**: canonical reps, multiplicities, and normalization checks. 6. **Numerical closures**: $S^{(r)}$ for $r \in \{0,1,2\}$ on the observable basis.

The bundle must verify end-to-end under the deterministic configuration.

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6) Complexity & Throughput Expectations

- **R96 enumeration:** linear in $|U_{\mathrm{ref}}| = 2048$ per orbit, with orbit reduction dominating per class; overall cost dominated by class discovery and hashing.
- **C768 schedule:** $O(768 \cdot |\mathcal{F}|)$ for a chosen observable family $\mathcal{F}$.
- **Φ checks:** dominated by orbit partitioning and equivariance tests; practical speedups from Schreier–Sims-style caching of stabilizers and coset reps.

These bounds size batches in the Moonshine pipeline.

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7) Pass/Fail Gates (Enforced at Integration)

All three gates must pass for acceptance: 1. **C768**: empirical sums match closed forms (exactly for integer cases or within certified numeric tolerances); fairness counts hold. 2. **Φ-compatibility**: equivariance holds on a spanning set; Φ-transported anchor orbits partition G . 3. **R96**: total class count is 96 with multiplicities $\{2:32, 3:64\}$ under the fixed selector/normalizer and unity relabelings.

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8) Category-Theoretic & Holographic Framing

Let **TetRhythm** be the groupoid of six-level tetrahedral rhythms with morphisms the boundary-equivariant maps. Let **MoonshineBoundary** be the category of boundary labelings with U_{ref} -equivariant morphisms. The Φ-map specifies a functor

$$\mathbf{\Phi} : \mathbf{TetRhythm} \longrightarrow \mathbf{MoonshineBoundary},$$

whose object component is $(A \times \mathbb{Z}_2^{10},\ M_p, M_b, \iota) \mapsto (G, U_{\mathrm{ref}} \curvearrowright G)$ and whose arrow component sends bulk-equivariant maps to boundary-equivariant maps. **Integrability** is the statement that $\mathbf{\Phi}$ is compatible with orbit decompositions (naturality with respect to the U_{ref} action) and preserves the C768 closure structure. The R96 canonicalization forms a **choice of section** of the orbit projection functor, fixed by the audit configuration.

Exploratory bridge. Within the Hologram APEX paradigm, Φ can be lifted to a monoidal functor that intertwines the Π-kernel and the Multiplicity Runtime, treating the 11 toggles as a graded object whose degree filtration matches the six anchor levels; this justifies the six-fold anchoring as a holographic boundary of a 6-simple (tetrahedral) bulk. (This bridge is speculative but mathematically consistent with the equivariance constraints.)

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9) Reference Algorithms (Deterministic Pseudocode)

9.1 Subgroup Freeness & Tiling

```
for a in 0..5:
  seen = empty set
  for toggle in 0..(2^11-1):
    u = toggle_to_generator_word(toggle)
    x = action(u, anchor[a])
    assert x not in seen
    seen.add(x)
  assert |seen| == 2048
accum = disjoint_union_of_all_seen_sets()
assert |accum| == 12288 and sets_are_pairwise_disjoint()
```

9.2 C768 Schedule & Closures

```
S0, S1, S2 = 0, 0, 0
for t in 0..767:
  (p,b) = (t mod 48, t mod 256)
  val = d + alpha[p] + gamma[b]
  S0 += val
  S1 += val * t
  S2 += val * t*t
# compare S0,S1,S2 to closed forms; also tally per-page and per-byte counts
```

9.3 Φ-Equivariance

```
# M_p, M_b in F2, iota: A->F2^6
for each generator u in U_ref:
  e = bulk_toggle[u]  # element of Z_2^10
  for a in A:
    for z in sample_bulk_states:  # spanning set
      lhs = Phi(a, z XOR e)
      rhs = action(u, Phi(a, z))
```

```
    assert lhs == rhs
# verify that Phi-transported anchor orbits partition G
```

9.4 R96 Canonicalization

```
def canonical_rep(orbit):
  best = None
  for word in gray_traverse_11bit():
    x = apply(word, orbit.seed)
    label = normalize(label_of(x))   # pair-normalized unity
    h = SHA256(salt || serialize(label))
    key = (h, serialize(label), word)
    if best is None or key < best.key:
      best = (key, x)
  return best.x

classes = set()
for orbit in enumerate_orbits(G, U_ref):
  rep = canonical_rep(orbit)
  classes.add(rep)
assert |classes| == 96 and multiplicities == {2:32, 3:64}
```

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10) Worked Targets for Quick Verification

For the constant observable $f \equiv 1$ on C768:

$$S^{(0)} = 768, \qquad S^{(1)} = 294{,}528, \qquad S^{(2)} = 150{,}700{,}160.$$

These serve as immediate regression tests for the schedule implementation and sum accumulation.

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11) Integration Checklist (Pass/Fail)

- [ ] **Boundary:** G , U_{ref} (order 2048), anchors S , freeness, 6×2048 tiling.
- [ ] **Schedule:** σ_{CFS} implemented; fairness counts; $S^{(r)}$ closures validated.
- [ ] **Φ:** matrices M_p, M_b , anchor embedding ι , equivariance table $u \mapsto e(u)$, Φ-tiling.
- [ ] **R96:** fixed generator order, salt; canonical classes = 96; multiplicities $\{2:32, 3:64\}$; normalization invariance.
- [ ] **Audit bundle:** all artifacts present; verifier script passes end-to-end.

Appendix A: Register↔Mod Map (implementation note)

- Pages use a 6-bit register; decoding to $\mathbb{Z}/48\mathbb{Z}$ must wrap values ≥ 48 by subtracting 48. Byte decoding is standard 8-bit to $\mathbb{Z}/256\mathbb{Z}$.
- The XOR action is applied on registers **before** decoding; this makes the action strictly $\mathbb{F}_2$ -linear at the bit level while respecting modular targets.

Appendix B: Determinism Policy

Every degree of freedom (generator order, salt, seeds, tie-breakers) is fixed in the manifest. All randomized routines are seeded from the manifest; all traversals use Gray-order to ensure reproducible orbit walks.

Appendix C (Exploratory): APEX Bridge to Π-Kernel & Multiplicity Runtime

Within Hologram APEX, interpret the 11 boundary toggles as a graded object $E = \bigoplus_{i=0}^{10} E_i$ over $\mathbb{F}_2$. The Φ-map lifts to a graded functor whose associated graded preserves C768 closures and R96 canonical sections. Anchors A give a 6-fold boundary decomposition that pairs naturally with a tetrahedral stratification of bulk degrees. This suggests a Π-kernel factorization of Φ and a multiplicity-preserving runtime on boundary orbits. (Exploratory, but consistent with §§1–4.)

End of Note